

Multisymplectic formalism for manifolds with boundaries

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1 Questions

- How can we encode the boundary conditions at the level of the action?
- What is the geometry describing the dynamics in manifolds with boundaries?

2 Relative (co)homology

On a manifold without boundary, we have the following formulation of Stokes’ theorem:

$$\int_{\mathcal{M}} d\alpha = 0 \tag{1}$$

When the boundary of the manifold on which we are working is non-zero, Stokes’ theorem takes the following form:

$$\int_{\mathcal{M}} d\alpha = \int_{\partial M} j^* \alpha \tag{2}$$

where $j : \partial\mathcal{M} \hookrightarrow \mathcal{M}$ is the inclusion.

Moreover, since the boundary of a manifold has no boundary (because $\partial^2\mathcal{M} = \phi$), Stokes' theorem becomes:

$$\int_{\partial\mathcal{M}} d\beta = 0 \quad (3)$$

We can write these two equations on the same side of the equality to obtain:

$$\int_{\mathcal{M}} d\alpha - \int_{\partial\mathcal{M}} j^*\alpha + \int_{\partial\mathcal{M}} d\beta = 0 \quad (4)$$

Let us introduce the following notation: (ω, η) is a differential form on $(\mathcal{M}, \mathcal{N})$ if ω is a differential form on $\mathcal{M}$ and η is a differential form on $\mathcal{N}$. We define the integral over this pair as:

$$\int_{(\mathcal{M}, \mathcal{N})} (\omega, \eta) = \int_{\mathcal{M}} \omega - \int_{\mathcal{N}} \eta \quad (5)$$

in such a way that we can simultaneously write the two Stokes theorems on the manifold and its boundary as:

$$\int_{(\mathcal{M}, \partial\mathcal{M})} (d\alpha, j^*\alpha - d\beta) = 0 \quad (6)$$

This way of simultaneously writing Stokes' theorems on the manifold and its boundary resembles what happens in the case where the manifold has no boundary, as in equation (1), but with a slightly different structure. Margalef's idea is to define the exterior derivative of the form (ω, η) as:

$$\underline{d}(\omega, \eta) = (d\omega, j^*\omega - d\eta) \quad (7)$$

so that Stokes' theorem with boundary can be written as:

$$\int_{(\mathcal{M}, \partial\mathcal{M})} \underline{d}(\alpha, \beta) = 0 \quad (8)$$

With this generalization of the exterior derivative, the pair $(\mathcal{M}, \partial\mathcal{M})$ behaves like a manifold without boundary with respect to this new (co)homology. One can check that the operation $\underline{d}$ is indeed an exterior derivative:

$$\begin{aligned} \underline{d}^2(\omega, \eta) &= \underline{d}(d\omega, j^*\omega - d\eta) \\ &= (d^2\omega, j^*(d\omega) - d(j^*\omega - d\eta)) \\ &= (0, 0) \end{aligned} \quad (9)$$

Furthermore, this cohomology defines a homology on the pair of manifolds $(\mathcal{M}, \mathcal{N})$, with $\mathcal{N} \subset \mathcal{M}$, given by:

$$\underline{\partial}(\mathcal{M}, \mathcal{N}) = (\partial\mathcal{M}/\mathcal{N}, -\partial\mathcal{N}) \quad (10)$$

This (co)homology is called relative.

We can verify that the relative exterior derivative on pairs of forms $\underline{d}$ is dual to the relative boundary on pairs of spaces $\underline{\partial}$ by proving Stokes' theorem with boundary.

Theorem (Stokes with boundaries):

$$\int_{(\mathcal{M}, \mathcal{N})} \underline{d}(\alpha, \beta) = \int_{\underline{\partial}(\mathcal{M}, \mathcal{N})} (\alpha, \beta) \quad (11)$$

Proof:

$$\begin{aligned}
\int_{(\mathcal{M}, \mathcal{N})} \underline{d}(\alpha, \beta) &= \int_{(\mathcal{M}, \mathcal{N})} (d\alpha, j^* \alpha - d\beta) \\
&= \int_{\mathcal{M}} d\alpha - \int_{\mathcal{N}} (j^* \alpha - d\beta) \\
&= \int_{\partial \mathcal{M}} \alpha - \int_{\mathcal{N}} j^* \alpha + \int_{\partial \mathcal{N}} \beta \\
&= \int_{\partial \mathcal{M}/\mathcal{N}} \alpha + \int_{\partial \mathcal{N}} \beta \\
&= \int_{(\partial \mathcal{M}/\mathcal{N}, -\partial \mathcal{N})} (\alpha, \beta) \\
&= \int_{\underline{\partial}(\mathcal{M}, \mathcal{N})} (\alpha, \beta)
\end{aligned} \tag{12}$$

□

Thus, the boundary of the boundary is also zero:

$$\begin{aligned}
\underline{\partial}^2(\mathcal{M}, \mathcal{N}) &= \underline{\partial}(\partial \mathcal{M} - \mathcal{N}, -\partial \mathcal{N}) \\
&= (\partial(\partial \mathcal{M} - \mathcal{N}) + \partial \mathcal{N}, \partial^2 \mathcal{N}) \\
&= (\phi, \phi)
\end{aligned} \tag{13}$$

However, in the case of a manifold with boundary, such as the one we are studying, the relative boundary is:

$$\begin{aligned}
\underline{\partial}(\mathcal{M}, \partial \mathcal{M}) &= (\partial \mathcal{M}/\partial \mathcal{M}, \partial^2 \mathcal{M}) \\
&= (\phi, \phi)
\end{aligned} \tag{14}$$

That is, with this homology, a manifold together with its boundary behaves as if it had no boundary with respect to this cohomology. This is why, in the Stokes theorem of equation (8), zero appears on the right-hand side of the equation.

On this pair one can define the usual operations, in addition to the exterior derivative:

$$\underline{\iota}_V(\omega, \eta) = (\iota_V \omega, -\iota_{\overline{V}} \eta), \quad \underline{\mathcal{L}}_V = (\mathcal{L}_V \omega, \mathcal{L}_{\overline{V}} \eta), \quad \text{where; } \overline{V} \equiv V|_{\partial \mathcal{M}} \tag{15}$$

$$(\omega, \eta) \underline{\wedge} (\xi, \theta) = \left(\omega \wedge \xi, \frac{(-1)^{n+1}}{2} j^* \omega \wedge \theta + \frac{1}{2} \eta \wedge j^* \xi \right) \tag{16}$$

3 General variational problem

3.1 Geometric construction

We have a fiber bundle $\pi : E \rightarrow \mathcal{M}$ and an inclusion $j : \partial \mathcal{M} \rightarrow \mathcal{M}$.

From this bundle and the inclusion, we can construct the pullback bundle $\pi_{\partial \mathcal{M}} : j^* E_{\partial \mathcal{M}} \rightarrow \partial \mathcal{M}$. Since j is an inclusion, the pullback bundle is canonically isomorphic to the restriction, i.e., $j^* E_{\partial \mathcal{M}} \cong E|_{\partial \mathcal{M}}$.

Given a section $\psi \in \Gamma(\mathcal{M}, E)$, there exists a unique section $\psi|_{\partial \mathcal{M}} \in \Gamma(\partial \mathcal{M}, E|_{\partial \mathcal{M}})$ such that $\psi \circ j = j_E \circ \psi|_{\partial \mathcal{M}}$

The following diagram is commutative:

$$\begin{array}{ccc}
E & \xleftarrow{j_E} & E_{\partial \mathcal{M}} \\
\psi \nearrow \downarrow \pi & & \pi_{\partial \mathcal{M}} \downarrow \nearrow \psi|_{\partial \mathcal{M}} \\
\mathcal{M} & \xleftarrow{j} & \partial \mathcal{M}
\end{array} \tag{17}$$

In the bulk, we of course work with the complex of differential forms on $\mathcal{M}$ up to degree n :

$$0 \xrightarrow{d} \Omega^0(\mathcal{M}) \xrightarrow{d} \Omega^1(\mathcal{M}) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{n-1}(\mathcal{M}) \xrightarrow{d} \Omega^n(\mathcal{M}) \xrightarrow{d} 0 \quad (18)$$

On the boundary we will also work with differential forms on $\mathcal{M}$ with support on $\partial\mathcal{M}$ (that is, differential forms of the form $f_{\mu_1 \dots \mu_k}(x^m) dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k}$), but only up to degree $n-1$. The complex is therefore:

$$0 \xrightarrow{d} \Omega^0(\partial\mathcal{M}) \xrightarrow{d} \Omega^1(\partial\mathcal{M}) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{n-2}(\partial\mathcal{M}) \xrightarrow{d} \Omega^{n-1}(\partial\mathcal{M}) \xrightarrow{d} 0 \quad (19)$$

Therefore, the differential complex of relative forms in which we will work is:

$$0 \xrightarrow{d} \Omega^0(\mathcal{M}) \xrightarrow{d} \Omega^1(\mathcal{M}) \oplus \Omega^0(\partial\mathcal{M}) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{n-1}(\mathcal{M}) \oplus \Omega^{n-2}(\partial\mathcal{M}) \xrightarrow{d} \Omega^n(\mathcal{M}) \oplus \Omega^{n-1}(\partial\mathcal{M}) \xrightarrow{d} 0 \quad (20)$$

Let $(\alpha, \beta) \in \Omega^k(E) \oplus \Omega^{k-1}(E|_{\partial\mathcal{M}})$. Given a section $\psi : \mathcal{M} \rightarrow E$, we can define the relative pullback as the relative pair formed by the pullbacks in the bulk and on the boundary:

$$\Psi^*(\alpha, \beta) = (\psi^*\alpha, \psi|_{\partial\mathcal{M}}^*\beta).$$

3.2 Variational problem and equations of motion

General variational problem on a manifold with boundary

Let $\pi : E \rightarrow \mathcal{M}$ be a fiber bundle and let $j : \partial\mathcal{M} \rightarrow \mathcal{M}$ denote the inclusion of the boundary. Let $(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) \in \Omega^n(E) \oplus \Omega^{n-1}(E|_{\partial\mathcal{M}})$.

We define the following functional on the space of smooth sections:

$$S[\psi] = \int_{(\mathcal{M}, \partial\mathcal{M})} \Psi^*(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) \quad (21)$$

Find the section ψ that minimizes this functional.

To obtain the equations that determine ψ , we must first prove a modified version of the fundamental lemma of the calculus of variations that includes boundaries. The modified version of the lemma is the following.

Lemma (fundamental lemma of the calculus of variations with submanifolds): Let $S \subseteq \mathcal{M}$ be a compact subset of a manifold $\mathcal{M}$ and let $\mathcal{N} \subseteq \mathcal{M}$ be a subset with empty interior. Let $\alpha \in \Omega^m(\mathcal{M})$ and $\beta \in \Omega^n(\mathcal{N})$. If

$$\int_S h\alpha + \int_{S \cap \mathcal{N}} h|_{\mathcal{N}}\beta = 0 \quad \forall h : \mathcal{M} \rightarrow \mathbb{R} \quad (22)$$

then it follows that $\alpha(x) = 0$ and $\beta(x) = 0$ on their respective domains.

Proof: If the integral vanishes for any function, then in particular it vanishes for a function h whose compact support is disjoint from $\mathcal{N}$. In that case the second integral vanishes, and therefore

$$\int_S h\alpha = 0, \quad \forall h : \text{supp}(h) = S \setminus \mathcal{N}. \quad (23)$$

By the fundamental lemma of the calculus of variations, α vanishes at every point of $S \setminus \mathcal{N}$. Since α is a continuous differential form, it must also vanish on S , including $\mathcal{N}$.

Finally, since $\alpha(x) = 0$ on all of S , we are left with

$$\int_{S \cap \mathcal{N}} h|_{\mathcal{N}}\beta = 0, \quad \forall h|_{\mathcal{N}}, \quad (24)$$

and again by the fundamental lemma of the calculus of variations we conclude that $\beta(x) = 0$ for all $x \in S \cap \mathcal{N}$.

□

Specializing to $\mathcal{N} = \partial\mathcal{M}$ and rewriting it using the notation introduced previously, we obtain that if

$$\int_{(\mathcal{M}, \partial\mathcal{M})} h(\alpha, \beta) = 0, \quad \forall h \in C^\infty(\mathcal{M}), \quad (25)$$

then $(\alpha, \beta) = 0$.

Once this lemma has been established, we can write the differential equations whose solutions minimize the action functional.

Theorem. The section ψ that minimizes the functional $S[\psi]$ for arbitrary variations satisfies

$$\Psi^* \iota_X d(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) = 0$$

for any vertical vector field X .

Proof: Let ψ_s be a smooth one-parameter family of sections and denote $\psi \equiv \psi_0$. For each point $p \in \mathcal{M}$, the curve $s \mapsto \psi_s(p)$ has a tangent vector

$$V(p) \equiv \partial_s \psi_s(p)|_{s=0},$$

which is vertical.

Choose any extension X of V such that $X|_{\psi_0(\mathcal{M})} = V$. One can show that

$$\frac{d}{ds}(\psi_s^* \Theta_{\mathcal{M}})|_{s=0} = \psi_0^*(\mathcal{L}_X \Theta_{\mathcal{M}}) \quad (26)$$

and that this equality does not depend on the chosen extension of V . Indeed, if Y is another extension satisfying $Y|_{\psi_0(\mathcal{M})} = 0$, then

$$\psi_0^*(\mathcal{L}_Y \Theta_{\mathcal{M}}) = \psi_0^*(d(\iota_Y \Theta_{\mathcal{M}}) + \iota_Y d\Theta_{\mathcal{M}}) = 0,$$

since $\iota_Y \Theta_{\mathcal{M}}$ and $\iota_Y d\Theta_{\mathcal{M}}$ vanish at points where $Y = 0$. Hence any two extensions yield the same pullback.

Therefore

$$\begin{aligned} \delta S[\phi] &= \frac{d}{ds} \left(\int_{\mathcal{M}} \psi_s^* \Theta_{\mathcal{M}} - \int_{\partial\mathcal{M}} \psi_s|_{\partial\mathcal{M}}^* \Theta_{\partial\mathcal{M}} \right) \Big|_{s=0} \\ &= \int_{\mathcal{M}} \psi^* \mathcal{L}_X \Theta_{\mathcal{M}} - \int_{\partial\mathcal{M}} \psi|_{\partial\mathcal{M}}^* \mathcal{L}_{X|_{\partial\mathcal{M}}} \Theta_{\partial\mathcal{M}} \\ &= \int_{\mathcal{M}} \psi^* (d\iota_X \Theta_{\mathcal{M}} + \iota_X d\Theta_{\mathcal{M}}) - \int_{\partial\mathcal{M}} \psi|_{\partial\mathcal{M}}^* (d\iota_{X|_{\partial\mathcal{M}}} \Theta_{\partial\mathcal{M}} + \iota_{X|_{\partial\mathcal{M}}} d\Theta_{\partial\mathcal{M}}) \\ &= \int_{\mathcal{M}} \psi^* \iota_X d\Theta_{\mathcal{M}} - \int_{\partial\mathcal{M}} \psi|_{\partial\mathcal{M}}^* (\iota_{X|_{\partial\mathcal{M}}} d\Theta_{\partial\mathcal{M}} - j^* \iota_X \Theta_{\mathcal{M}}) \\ &= \int_{(\mathcal{M}, \partial\mathcal{M})} \Psi^* (\iota_X d\Theta_{\mathcal{M}}, \iota_{X|_{\partial\mathcal{M}}} d\Theta_{\partial\mathcal{M}} - j^* \iota_X \Theta_{\mathcal{M}}). \end{aligned} \quad (27)$$

Now the vector field X , which generates the parametric transformation ψ_s , is vertical. Hence $X|_{\partial\mathcal{M}} = X$, and the contraction can be taken outside the pullback along the inclusion:

$$\begin{aligned} \delta S[\phi] &= \int_{(\mathcal{M}, \partial\mathcal{M})} \Psi^* \iota_X (d\Theta_{\mathcal{M}}, j^* \Theta_{\mathcal{M}} - d\Theta_{\partial\mathcal{M}}) \\ &= \int_{(\mathcal{M}, \partial\mathcal{M})} \Psi^* \iota_X d(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}). \end{aligned} \quad (28)$$

Taking the action to be stationary, $\delta S = 0$, we obtain

$$\int_{(\mathcal{M}, \partial\mathcal{M})} \Psi^* \iota_X d(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) = 0. \quad (29)$$

By assumption of the theorem, this holds for any vector field X . In particular it holds for any vector field of the form $X = hY$, where h is an arbitrary function:

$$\int_{(\mathcal{M}, \partial\mathcal{M})} \Psi^* h \iota_Y \underline{d}(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) = 0. \quad (30)$$

By the fundamental lemma of the calculus of variations with boundary, since the integral vanishes for any function h , the factor multiplying h must vanish. We thus obtain the equations of motion:

$$\Psi^* \iota_X \underline{d}(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) = 0 \quad \forall X \text{ vertical}. \quad (31)$$

This proves the theorem. $\square$

As usual, we define the relative multisymplectic form

$$\Omega = (\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}})$$

as the relative exterior derivative of the forms defining the functional,

$$\Omega \equiv \underline{d}(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}).$$

Hence the section ψ minimizing the functional satisfies

$$\Psi^* \iota_X \Omega = 0 \quad \forall X \text{ vertical}. \quad (32)$$

For each variational principle it is natural to define its (pre)multiplesymplectic form

$$\Omega = (\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}) \equiv \underline{d}(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}).$$

Since the relative exterior derivative also satisfies $\underline{d}^2 = 0$, the manifold with boundary $(\mathcal{M}, \partial\mathcal{M})$ is endowed with an $(n+1, n)$ -form

$$\Omega = (\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}})$$

satisfying $\underline{d}\Omega = (0, 0)$. This motivates the following definition of a multisymplectic manifold with boundary.

Definition (multisymplectic manifold with boundary). A manifold with boundary $(\mathcal{M}, \partial\mathcal{M})$ is called n -multisymplectic if and only if it is endowed with an $(n+1, n)$ multisymplectic form $\Omega = (\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}})$ such that $\underline{d}\Omega = (0, 0)$ and such that $\iota_V \Omega = 0$ if and only if $V = 0$. If only the first condition holds, the manifold with boundary is called *premultisymplectic*.

In this way, working with the multisymplectic formalism on a manifold with boundary is analogous to working with the multisymplectic formalism on a manifold without boundary, but replacing the usual cohomology with relative cohomology. All properties of the multisymplectic formalism extend naturally to manifolds with boundary simply by replacing differential forms with relative differential forms and replacing the usual exterior derivative with the relative exterior derivative.

4 Hamiltonian forms, graded Poisson brackets and conserved charges

Definition (locally Hamiltonian multivector field). A multivector field X is called *locally Hamiltonian* if $\iota_X \Omega$ is a closed form with respect to the relative exterior derivative $\underline{d}$.

Theorem. If X is a locally Hamiltonian p -multivector field, then

$$\underline{\mathcal{L}}_X \Omega = 0.$$

Proof.

$$\underline{\mathcal{L}}_X \Omega = \iota_X \underline{d}\Omega - (-1)^p \underline{d}\iota_X \Omega. \quad (33)$$

The first term vanishes because the multisymplectic form is closed. The second term vanishes because $\iota_X \Omega$ is closed. Therefore $\underline{\mathcal{L}}_X \Omega = 0$. $\square$

This means that locally Hamiltonian vectors are precisely those that preserve the geometric structure of the variational principle both in the bulk and on the boundary.

Definition (globally Hamiltonian multivector field). A multivector field X is called *globally Hamiltonian* if $\iota_X \Omega$ is exact. The pair

$$F = (F_{\mathcal{M}}, F_{\partial\mathcal{M}})$$

satisfying

$$\iota_X \Omega = dF$$

is called the Hamiltonian form associated with the multivector X .

Remark. Separating these equations yields

$$\iota_X \Omega_{\mathcal{M}} = dF_{\mathcal{M}}, \quad \iota_{\bar{X}} \Omega_{\partial\mathcal{M}} = dF_{\partial\mathcal{M}} - j^* F_{\mathcal{M}}. \quad (34)$$

Thus the relation between multivectors and Hamiltonian forms remains unchanged in the bulk, while on the boundary the bulk and boundary contributions become mixed.

Proposition. Every globally Hamiltonian multivector field is locally Hamiltonian.

Proof. Since $d^2 = 0$, if $\iota_X \Omega$ is exact then it is also closed. $\square$

Since Hamiltonian multivectors preserve the multisymplectic form Ω , they are interpreted as symmetries of the dynamics, both in the bulk and on the boundary.

Definition. Let X be the Hamiltonian vector associated with

$$F = (F_{\mathcal{M}}, F_{\partial\mathcal{M}})$$

and let Y be the Hamiltonian vector associated with

$$G = (G_{\mathcal{M}}, G_{\partial\mathcal{M}}).$$

We define the Poisson bracket $\{F, G\}$ as the observable associated with the Schouten–Nijenhuis bracket $[X, Y]$:

$$\iota_{[X, Y]}(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}) = d\{F, G\}. \quad (35)$$

Theorem. The graded Poisson bracket of $F = (F_{\mathcal{M}}, F_{\partial\mathcal{M}})$ and $G = (G_{\mathcal{M}}, G_{\partial\mathcal{M}})$ is, up to closed relative forms,

$$\{F, G\} = \iota_X \iota_Y (\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}). \quad (36)$$

This theorem will be proven later. Before that we establish several preliminary lemmas.

Proposition.

$$[U, V]|_{\partial\mathcal{M}} = [U|_{\partial\mathcal{M}}, V|_{\partial\mathcal{M}}].$$

Proof. A vector $X \in \mathfrak{X}(\mathcal{M})$ is tangent to $\mathcal{N} \subset \mathcal{M}$ if

$$X(f)|_{\mathcal{N}} = 0$$

for all functions f such that $f|_{\mathcal{N}} = 0$.

Suppose X and Y are tangent to $\mathcal{N}$. Then for every function f vanishing on $\mathcal{N}$,

$$X(f)|_{\mathcal{N}} = 0, \quad Y(f)|_{\mathcal{N}} = 0.$$

Hence

$$[X, Y](f)|_{\mathcal{N}} = X(Y(f))|_{\mathcal{N}} - Y(X(f))|_{\mathcal{N}}. \quad (37)$$

Since $f|_{\mathcal{N}} = 0$, both $X(f)$ and $Y(f)$ vanish on $\mathcal{N}$, implying that $X(Y(f))$ and $Y(X(f))$ also vanish on $\mathcal{N}$. Therefore the Lie bracket of two tangent vectors is also tangent.

More generally, the Schouten–Nijenhuis bracket of two decomposable multivectors

$$U = U_1 \wedge \cdots \wedge U_u, \quad V = V_1 \wedge \cdots \wedge V_v$$

can be written as

$$[U, V] = \sum_{i=1}^u \sum_{j=1}^v [U_i, V_j] \wedge U_1 \wedge \cdots \wedge \tilde{U}_i \wedge \cdots \wedge U_u \wedge V_1 \wedge \cdots \wedge \tilde{V}_j \wedge \cdots \wedge V_v, \quad (38)$$

where $\tilde{U}_i$ indicates that the term is omitted.

Since restriction commutes with both the Lie bracket and the wedge product, the restriction of the Schouten–Nijenhuis bracket equals the bracket of the restricted multivectors. By linearity this extends to general multivectors, proving the statement. $\square$

Property. Let U and V be a u -multivector and a v -multivector, respectively. Then

$$\iota_{[U,V]}(\alpha, \beta) = [\mathcal{L}_U, \iota_V](\alpha, \beta). \quad (39)$$

Lemma. If U and V are Hamiltonian multivector fields, then

$$\iota_{[U,V]}(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}) = \underline{d}\iota_U \iota_V(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}). \quad (40)$$

Proof. Using the previous identity and the fact that $\underline{d}\Omega = 0$ and $\underline{d}\iota_U \Omega = \underline{d}\iota_V \Omega = 0$, the result follows directly. $\square$

Proof of the theorem. By definition, $\underline{d}\{F, G\} = \iota_{[X,Y]}(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}})$. Using the lemma, $\iota_{[X,Y]}(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}) = \underline{d}\iota_X \iota_Y(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}})$. Hence $\underline{d}\{F, G\} = \underline{d}(\iota_X \iota_Y(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}))$, which implies that the Poisson bracket is, up to closed relative forms,

$$\{F, G\} = \{(F_{\mathcal{M}}, F_{\partial\mathcal{M}}), (G_{\mathcal{M}}, G_{\partial\mathcal{M}})\} = \iota_X \iota_Y(\Omega_{\mathcal{M}}, \Omega_{\partial\mathcal{M}}) = (\iota_X \iota_Y \Omega, \iota_X|_{\partial\mathcal{M}} \iota_Y|_{\partial\mathcal{M}} \omega) \quad (41)$$

Definition (conserved quantities with boundary). A relative form

$$F = (F_{\mathcal{M}}, F_{\partial\mathcal{M}})$$

is said to be conserved if for every solution ψ of the bulk and boundary equations of motion one has

$$\Psi^* \underline{d}F = (0, 0).$$

Definition (conserved charges). Let Σ be a p -submanifold of $\mathcal{M}$ and σ a $(p-1)$ -submanifold of $\partial\mathcal{M}$. Given a conserved $(p, p-1)$ -form F , its conserved charge is defined as

$$Q_F[\Sigma, \sigma] = \int_{(\Sigma, \sigma)} \Psi^* F = \int_{\Sigma} \psi^* F_{\mathcal{M}} - \int_{\sigma} \psi|_{\partial\mathcal{M}}^* F_{\partial\mathcal{M}}. \quad (42)$$

Theorem. Conserved charges are constants of motion.

If (Σ_1, σ_1) and (Σ_2, σ_2) enclose a relative volume (V, v) such that $\partial(V, v) = (\Sigma_1 \cup (-\Sigma_2), \sigma_1 \cup (-\sigma_2))$, then $Q_F[\Sigma_1, \sigma_1] = Q_F[\Sigma_2, \sigma_2]$.

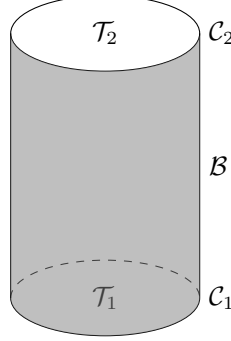
Proof.

$$\begin{aligned} Q_F[\Sigma_1, \sigma_1] - Q_F[\Sigma_2, \sigma_2] &= \int_{(\Sigma_1, \sigma_1)} \Psi^* F - \int_{(\Sigma_2, \sigma_2)} \Psi^* F \\ &= \int_{(\Sigma_1, \sigma_1)} \Psi^* F + \int_{(-\Sigma_2, -\sigma_2)} \Psi^* F \\ &= \int_{(\Sigma_1 - \Sigma_2, \sigma_1 - \sigma_2)} \Psi^* F \\ &= \int_{\partial(V, S)} \Psi^* F \\ &= \int_{(V, S)} \Psi^* \underline{d}F \\ &= 0 \end{aligned} \quad (43)$$

from which it follows that $Q_F[\Sigma_1, \sigma_1] = Q_F[\Sigma_2, \sigma_2]$, as we wanted to prove. In the step to the penultimate line we have used Stokes' theorem for relative cohomology. $\square$

A relevant example in which this theorem holds is when the manifold $\mathcal{M}$ has the topology $\mathcal{N} \times [t_1, t_2]$. The boundary of the manifold is formed by the lower cap $\mathcal{T}_1 = \mathcal{N} \times \{t_1\}$ and the upper cap $\mathcal{T}_2 = \mathcal{N} \times \{t_2\}$,

the side of the manifold $\mathcal{B} = \partial\mathcal{N} \times [t_1, t_2]$, and the two circles forming the corners $\mathcal{C}_1 = \partial\mathcal{N} \times \{t_1\}$ and $\mathcal{C}_2 = \partial\mathcal{N} \times \{t_2\}$. If the surfaces Σ_1 and Σ_2 are chosen to be the lower and upper caps $\mathcal{T}_1$ and $\mathcal{T}_2$, and the varieties σ_1 and σ_2 are chosen as the circles $\mathcal{C}_1$ and $\mathcal{C}_2$ that form the corners of the manifold, one can check that the relative pair $(\mathcal{M}, \mathcal{B})$ has relative boundary $(\Sigma_1 \cup (-\Sigma_2), \sigma_1 \cup (-\sigma_2))$, following formula 10. Therefore, the conserved charge measured on $(\mathcal{T}_1, \mathcal{C}_1)$ is equal to the conserved charge measured on $(\mathcal{T}_2, \mathcal{C}_2)$.



The condition $\underline{d}(\Psi^*F) = 0$ is a conservation equation analogous to a continuity equation, but it has components both in the bulk and on the boundary. Let us consider the example of a relative $(n-1, n-2)$ -form. Such a pair is given, on solutions, by $\Psi^*F = (\alpha^\mu d^{n-1}x_\mu, \beta^{\mu\nu} d^{n-2}x_{\mu\nu})$. Saying that this differential form is conserved both in the bulk and on the boundary is equivalent to requiring that the following condition holds for the pair:

$$\Psi^*\underline{d}F = (\partial_\mu \alpha^\mu d^n x, (\alpha^\mu - \partial_\nu \beta^{\nu\mu} + \partial_\nu \beta^{\mu\nu}) d^{n-1}x_\mu) = 0 \quad (44)$$

Or, separating the two equations:

$$\partial_\mu \alpha^\mu = 0, \quad 2\partial_\mu \beta^{\mu\nu} = \alpha^\nu \quad (45)$$

If we take a foliation $t = \text{cte.}$ and rename $\alpha^0 = \rho$, $\alpha^i = J^i$, $\beta^{0i} = u^i$ and $\beta^{ij} = v^{ij}$, then we obtain the following continuity equations:

$$\frac{\partial \rho}{\partial t} + \partial_i J^i = 0, \quad -2\partial_i u^i = \rho, \quad 2\frac{\partial u^i}{\partial t} - 2\partial_j v^{ij} = J^i \quad (46)$$

This same exercise can be performed with other $(k, k-1)$ -forms, leading to continuity equations for conserved charges of different orders.

Theorem (Noether): If X_F is a globally Hamiltonian vector field, then its associated relative form $F = (F_{\mathcal{M}}, F_{\partial\mathcal{M}})$ is a conserved quantity with boundary.

Proof: We can use that X_F is a globally Hamiltonian vector field to write $\Psi^*\underline{d}F = \Psi^*\iota_{X_F}\Omega$. Now, using the equations of motion, $\Psi^*\iota_X\Omega = 0$ for any vector, and therefore $\Psi^*\underline{d}F = 0$.

Theorem: If $F = (F_{\mathcal{M}}, F_{\partial\mathcal{M}})$ and $G = (G_{\mathcal{M}}, G_{\partial\mathcal{M}})$ are two conserved $(n, n-1)$ -forms with boundary associated with the globally Hamiltonian vectors X and Y , then the Poisson bracket $\{F, G\}$ is also a conserved quantity with boundary.

Proof: By definition of the Poisson bracket, $\underline{d}\{F, G\} = \iota_{[X, Y]}\Omega$. Taking the pullback on both sides, $\Psi^*\underline{d}\{F, G\} = \Psi^*\iota_{[X, Y]}\Omega$. But if X and Y are two vectors, then $[X, Y]$ is also a vector, and therefore the equations of motion hold $\Psi^*\iota_{[X, Y]}\Omega = 0$. Of course, this implies that $\Psi^*\underline{d}\{F, G\} = 0$. $\square$

5 Lagrangian formalism

5.1 Geometric Lagrangian construction

We have a fiber bundle $\pi : E \rightarrow \mathcal{M}$ and an inclusion $j : \mathcal{M} \rightarrow \partial\mathcal{M}$. Let $V_{\mathcal{M}}$ be the volume form on $\mathcal{M}$ and $V_{\partial\mathcal{M}}$ be the volume form on $\partial\mathcal{M}$. The normal 1-form is canonically defined by $\iota_Y V_{\mathcal{M}}|_{\partial\mathcal{M}} = n(Y)V_{\partial\mathcal{M}}$. The normal unitary vector, satisfying $n^\mu n_\mu = 1$ is, however, non-unique. In fact, it requires the choice of a horizontal splitting of $T\mathcal{M} = j^*T(\partial\mathcal{M}) \oplus H$, i.e., a connection.

From this bundle and the inclusion, we can construct the pullback bundle $\pi_{\partial\mathcal{M}} : j^*E_{\partial\mathcal{M}} \rightarrow \partial\mathcal{M}$. Since j is an inclusion, the pullback bundle is canonically isomorphic to the restriction, i.e., $j^*E_{\partial\mathcal{M}} \cong E|_{\partial\mathcal{M}}$.

On the other hand, from $\pi : E \rightarrow \mathcal{M}$ we can canonically construct the jet bundle $\pi_1 : J^1E \rightarrow E$ as the space of sections that agree up to first order, that is, given a point $p \in \mathcal{M}$, they have the same value and the same derivatives at that point. The local coordinates of $J^1\pi$ are (x^μ, y^i, y_μ^i) . As in the previous case, from this bundle and the inclusion we can construct the pullback of the jet bundle $\pi_{1\partial\mathcal{M}} : j^*J^1E \rightarrow E_{\partial\mathcal{M}}$. Again, since j is an inclusion, we have $j^*J^1E \cong J^1E|_{\partial\mathcal{M}}$. Note that the coordinates of $J^1E|_{\partial\mathcal{M}}$ are $\{y^i, y_\mu^i, x^\mu\}$, and therefore $J^1E|_{\partial\mathcal{M}}$ is not the jet bundle of $E_{\partial\mathcal{M}}$.

Given a section $j^1\phi \in \Gamma(\mathcal{M}, J^1E)$, there exists a unique section $j^1\phi|_{\partial\mathcal{M}} \in \Gamma(\partial\mathcal{M}, J^1E|_{\partial\mathcal{M}})$ such that $j^1\phi \circ j = j_{1E} \circ j^1\phi|_{\partial\mathcal{M}}$. A section $j^1\phi(x^\mu) = (y^i(x^\mu), y_\mu^i(x^\mu), x^\mu)$ is said to be holonomic if $y_\mu^i(x^\mu) = \partial_\mu y^i(x^\mu)$.

The following diagram is commutative.

$$\begin{array}{ccc}
 J^1E & \xleftarrow{j_{1E}} & J^1E|_{\partial\mathcal{M}} \\
 \downarrow \pi_1 & & \downarrow \pi_{1\partial\mathcal{M}} \\
 E & \xleftarrow{j_E} & E_{\partial\mathcal{M}} \\
 \downarrow \pi & & \downarrow \pi_{\partial\mathcal{M}} \\
 \mathcal{M} & \xleftarrow{j} & \partial\mathcal{M}
 \end{array}
 \quad \begin{array}{l}
 \nearrow j^1\phi \\
 \searrow j^1\phi|_{\partial\mathcal{M}}
 \end{array}
 \quad (47)$$

In the bulk, of course, we work on the complex of differential forms on $\mathcal{M}$ up to degree n :

$$0 \xrightarrow{d} \Omega^0(\mathcal{M}) \xrightarrow{d} \Omega^1(\mathcal{M}) \xrightarrow{d} \dots \xrightarrow{d} \Omega^{n-1}(\mathcal{M}) \xrightarrow{d} \Omega^n(\mathcal{M}) \xrightarrow{d} 0 \quad (48)$$

On the boundary we will also work with differential forms on $\mathcal{M}$ with support on $\partial\mathcal{M}$ (that is, differential forms of the type $f_{\mu_1 \dots \mu_k}(x^\mu) dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}$), but only up to degree $n-1$. The complex is therefore:

$$0 \xrightarrow{d} \Omega^0(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} \Omega^1(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} \dots \xrightarrow{d} \Omega^{n-2}(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} \Omega^{n-1}(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} 0 \quad (49)$$

This means that, although we might encounter some form $f^\mu(x^\mu) d^{n-1}x_\mu$ with support on $\partial\mathcal{M}$, and although typically this differential form on $\mathcal{M}$ could have an exterior derivative that is an n -form, in this paper we will truncate this differential complex so that these n -forms with support on $\partial\mathcal{M}$ do not exist. This is because, in the end, we will always have to take the pullback to $\partial\mathcal{M}$ (for instance when integrating), and the pullback of n -forms onto $\partial\mathcal{M}$ will always be zero, so it will be more productive to truncate it from the start. Note that the inclusion $j : \partial\mathcal{M} \rightarrow \mathcal{M}$ allows us to define the restriction $j^* : \Omega^k(\mathcal{M}) \rightarrow \Omega^k(\mathcal{M})|_{\partial\mathcal{M}}$ given simply by restricting the coefficients of the differential form

$$j^*(f_{\mu_1 \dots \mu_k}(x^\mu) dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}) = f_{\mu_1 \dots \mu_k}(x^\mu) dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k},$$

except for the n -forms, where the restriction is zero due to the truncation of the differential complex.

Finally, the relative complex in which we will be working is:

$$0 \xrightarrow{d} \Omega^0(\mathcal{M}) \xrightarrow{d} \Omega^1(\mathcal{M}) \oplus \Omega^0(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} \dots \xrightarrow{d} \Omega^{n-1}(\mathcal{M}) \oplus \Omega^{n-2}(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} \Omega^n(\mathcal{M}) \oplus \Omega^{n-1}(\mathcal{M})|_{\partial\mathcal{M}} \xrightarrow{d} 0 \quad (50)$$

5.2 Equations of motion under the integral sign

In the Lagrangian formalism, unlike the general case, we will not look for an arbitrary section that minimizes the functional, but rather we will look for the minimum of that functional inside a subspace of all sections, namely those sections that are holonomic, i.e., those sections of J^1E that satisfy that the velocity y_μ^i is the derivative of the field $\partial_\mu y^i$.

Since the manifold $\mathcal{M}$ has dimension n , its boundary will have dimension $n-1$. Therefore the actions will result from integrating an n -form $\mathcal{L} d^n x$ in the bulk and from integrating an $(n-1)$ -form $\ell^\mu d^{n-1}x_\mu$ on the boundary. The variational problem, in the Lagrangian formalism, is therefore the following:

Lagrangian variational problem on a manifold with boundary

Let $\pi : J^1 E \rightarrow \mathcal{M}$ be a bundle and $j : \partial \mathcal{M} \rightarrow \mathcal{M}$ its boundary inclusion. Let $\mathcal{L} \in C^\infty(J^1 E)$ and $\ell^\mu \in C^\infty(J^1 E|_{\partial \mathcal{M}})$, with which we build the relative pair $(\mathcal{L}d^n x, -\ell^\mu d^{n-1} x_\mu) \in \Omega^n(J^1 E) \oplus \Omega^{n-1}(J^1 E|_{\partial \mathcal{M}})$. We define the action as the following functional on the set of smooth holonomic sections:

$$\begin{aligned} S[y^i] &= \int_{(\mathcal{M}, \partial \mathcal{M})} (J^1 \phi)^* (\mathcal{L}d^n x, -\ell^\mu d^{n-1} x_\mu) \\ &= \int_{\mathcal{M}} \mathcal{L}(y^i, \partial_\mu y^i, x^\mu) d^n x + \int_{\partial \mathcal{M}} \ell^\mu(y^i, \partial_\mu y^i, x^\mu) d^{n-1} x_\mu \end{aligned} \quad (51)$$

Find the holonomic section $(j^1 \phi)(x^\mu) = (y^i(x^\mu), \partial_\mu y^i(x^\mu), x^\mu)$ that minimizes the action.

Because here we are restricting to holonomic sections, the admissible variations are not arbitrary but only a subset of them, those that preserve holonomicity. Therefore the equations we previously obtained for a general variational problem will not be valid here, although those general equations will be important later when we compute the Poincaré–Cartan form in the Lagrangian formalism.

Now, to find the equations of motion, both in the bulk and on the boundary, we take a variation of the fields that respects the holonomicity of our system, so that the fields transform as:

$$\begin{aligned} \phi &\mapsto \phi + \varepsilon \\ \partial_\mu \phi &\mapsto \partial_\mu \phi + \partial_\mu \varepsilon \end{aligned} \quad (52)$$

The variation of the action under these variations can be computed as follows:

$$\begin{aligned} \delta S[y^i] &= \int_{\mathcal{M}} \left(\frac{\partial \mathcal{L}}{\partial y^i} \varepsilon^i + \frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} \partial_\mu \varepsilon^i \right) d^n x \\ &\quad + \int_{\partial \mathcal{M}} \left(\frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i + \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} \partial_\nu \varepsilon^i \right) d^{n-1} x_\mu \end{aligned} \quad (53)$$

In the bulk we can integrate by parts, obtaining:

$$\begin{aligned} \delta S &= \int_{\mathcal{M}} \left(\frac{\partial \mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} \right) \right) \varepsilon^i d^n x \\ &\quad + \int_{\partial \mathcal{M}} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} \varepsilon^i + \frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i + \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} \partial_\nu \varepsilon^i \right) d^{n-1} x_\mu \end{aligned} \quad (54)$$

On the boundary, however, the term inside the derivative cannot be fully integrated by parts to leave a component proportional to ε from which to obtain the equations of motion directly using the fundamental lemma of the calculus of variations. This is because terms of the form $n^\mu \partial_\mu \varepsilon$, proportional to the normal derivative of the variation at the boundary, cannot be integrated by parts. To see this more concretely, we can use adapted coordinates $x^\mu = \{x^0, \tilde{x}^i\}$, so that the boundary is given by $x^0 = \text{const.}$, and the normal vector to the hypersurface is $n^\mu = (1, 0, \dots, 0)$. The integral $\int \partial_0 \varepsilon d^{n-1} \tilde{x}$ cannot be integrated by parts, since the coordinate x^0 does not belong to the integration measure.

To keep under control the parts we can integrate by parts and those we cannot, we can split the derivative $\partial_\nu \varepsilon^i$ into its normal and tangential components to the boundary using the metric. For this, we introduce the projector onto the tangent plane of the boundary, defined by $P_\nu^\mu \equiv \delta_\nu^\mu - n^\mu n_\nu$. It is easy to check from this definition that $P_\nu^\mu P_\rho^\nu = P_\rho^\mu$ and that $P_\nu^\mu n^\nu = 0$. Thus the term $\partial_\nu \varepsilon^i$ can be rewritten as $(P_\nu^\mu + n_\nu n^\mu) \partial_\rho \varepsilon^i$, so we substitute:

$$\begin{aligned} \delta S &= \int_{\mathcal{M}} \left(\frac{\partial \mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} \right) \right) \varepsilon^i d^n x \\ &\quad + \int_{\partial \mathcal{M}} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} \varepsilon^i + \frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i + \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} P_\nu^\rho \partial_\rho \varepsilon^i \right) d^{n-1} x_\mu \\ &\quad + \int_{\partial \mathcal{M}} \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} n_\nu n^\rho \partial_\rho \varepsilon^i d^{n-1} x_\mu \end{aligned} \quad (55)$$

As argued, the third term of the last line can be integrated by parts. For this we use the computation of the following exterior derivative:

$$\begin{aligned}
d\left(\frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\varepsilon d^{n-2}x_{\mu\rho}\right) &= \left(\frac{d}{dx^\sigma}\left(\frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\right)\varepsilon + \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\partial_\sigma\varepsilon\right)dx^\sigma\wedge d^{n-2}x_{\mu\rho} \\
&= \left(\frac{d}{dx^\sigma}\left(\frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\right)\varepsilon + \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\partial_\sigma\varepsilon\right)(\delta_\mu^\sigma d^{n-1}x_\rho - \delta_\rho^\sigma d^{n-1}x_\mu) \\
&= \left(\frac{d}{dx^\rho}\left(\frac{\partial\ell^\rho}{\partial(\partial_\nu y^i)}P_\nu^\mu - \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\right)\right. \\
&\quad \left.+ \frac{\partial\ell^\rho}{\partial(\partial_\nu y^i)}P_\nu^\mu\partial_\rho\varepsilon - \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\partial_\rho\varepsilon\right)d^{n-1}x_\mu
\end{aligned} \tag{56}$$

The last term of the preceding expression is the same as the one appearing in the variation of the action, so we can solve for it in terms of the rest, thereby performing the integration by parts. Notice that the middle term does not contribute to the integral either, because $P_\nu^\mu d^{n-1}x_\mu$ will be proportional to $P_\nu^\mu n_\mu = 0$ when we take the pullback by the inclusion; therefore, integrating by parts in this manner leaves no derivative of ε . Applying Stokes' theorem, and using that the boundary of the boundary vanishes, what this calculation tells us is that the correct way to integrate by parts on the boundary (only tangential derivatives) is:

$$\int_{\partial\mathcal{M}} \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\partial_\rho\varepsilon^i d^{n-1}x_\mu = \int_{\partial\mathcal{M}} \frac{d}{dx^\rho}\left(\frac{\partial\ell^\rho}{\partial(\partial_\nu y^i)}P_\nu^\mu - \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\right)\varepsilon^i d^{n-1}x_\mu \tag{57}$$

Therefore the variation of the action is:

$$\begin{aligned}
\delta S &= \int_{\mathcal{M}} \left(\frac{\partial\mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu}\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu y^i)}\right)\right)\varepsilon^i d^n x \\
&\quad + \int_{\partial\mathcal{M}} \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu y^i)} + \frac{\partial\ell^\mu}{\partial y^i} + \frac{d}{dx^\rho}\left(\frac{\partial\ell^\rho}{\partial(\partial_\nu y^i)}P_\nu^\mu - \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}P_\nu^\rho\right)\right)\varepsilon^i d^{n-1}x_\mu \\
&\quad + \int_{\partial\mathcal{M}} \frac{\partial\ell^\mu}{\partial(\partial_\nu y^i)}n_\nu n^\rho\partial_\rho\varepsilon^i d^{n-1}x_\mu
\end{aligned} \tag{58}$$

Thus we have separated the variation of the action into three terms: one that affects only the bulk, another that is proportional to the variations on the boundary, and another that is proportional to the normal derivative of the variations on the boundary. We must now show, using methods similar to the fundamental lemma of the calculus of variations, that each of these terms vanishes separately in order for the variation of the action to be zero. The result we prove is the following:

Lemma (fundamental lemma of the calculus of variations with boundary): Let $\alpha \in C^\infty(\mathcal{M})$, $\beta \in C^\infty(\partial\mathcal{M})$ and $\gamma \in C^\infty(\partial\mathcal{M})$. If

$$\int_{\mathcal{M}} \alpha \varepsilon d^n x + \int_{\partial\mathcal{M}} \beta \varepsilon|_{\partial\mathcal{M}} d^{n-1}\tilde{x} + \int_{\partial\mathcal{M}} \gamma n^\mu\partial_\mu\varepsilon|_{\partial\mathcal{M}} d^{n-1}\tilde{x} = 0 \tag{59}$$

for every smooth function ε with compact support in $\mathcal{M} \cup \partial\mathcal{M}$, then $\alpha = 0$, $\beta = 0$ and $\gamma = 0$.

To prove it, we first need the following lemma:

Lemma (separation of normal derivatives): Let $\alpha, \beta \in C^\infty(\partial\mathcal{M})$. Then there exists a unique function $\varepsilon \in C^\infty(\mathcal{M})$ such that:

$$\varepsilon|_{\partial\mathcal{M}} = \alpha, \quad n^\mu\partial_\mu\varepsilon|_{\partial\mathcal{M}} = \beta. \tag{60}$$

Moreover, ε can be chosen with support in an arbitrarily small collar neighbourhood of $\partial\mathcal{M}$.

Proof: This follows from the collar neighbourhood theorems for manifolds with boundary: there exists a diffeomorphism $\partial\mathcal{M} \times (-\delta, 0] \cong U \subset \mathcal{M}$ that maps (x, t) to points at distance t from x towards the interior of the manifold along the flow of the normal vector. In these coordinates we construct $\varepsilon(x, t) = \alpha(x) + t\beta(x)\chi(t)$, where $\chi : (-\delta, 0] \rightarrow \mathbb{R}$ is a function supported in $(-\delta, 0]$ with $\chi(0) = 1$. Then ε is smooth in the collar, equals α at $t = 0$, and has normal derivative β at $t = 0$. One can extend ε by zero outside the collar to obtain a globally defined function. $\square$

Proof of the fundamental lemma: Since the integral vanishes for every compactly supported ε in $\mathcal{M} \cup \partial\mathcal{M}$, it also vanishes for every ε compactly supported in the interior $\mathcal{M} \setminus \partial\mathcal{M}$. This leaves only the first integral, i.e.

$$\int_{\mathcal{M}} \alpha \varepsilon d^n x = 0 \quad (61)$$

for every ε compactly supported in the interior of $\mathcal{M}$. By the usual fundamental lemma of the calculus of variations this implies $\alpha = 0$ on $\text{int}(\mathcal{M})$, and by continuity $\alpha = 0$ on $\mathcal{M} \cup \partial\mathcal{M}$.

Knowing that $\alpha = 0$, it remains to show that the other two terms vanish. For this we use the previous lemma, which allows us to prescribe the function and its normal derivative independently on the boundary. First choose a function ε with arbitrary boundary value but with $n^\mu \partial_\mu \varepsilon|_{\partial\mathcal{M}} = 0$. Then the fundamental lemma of the calculus of variations applied on the boundary implies $\beta = 0$. Conversely, choose a function ε that vanishes on the boundary but has an arbitrary normal derivative; applying the same argument shows $\gamma = 0$. $\square$

Hence the variational principle, which states that the action (51) must be stationary for any variation ε^i , implies that the coefficients in each of the integrals in (58) must vanish separately. Therefore we obtain the Euler–Lagrange equations as equations of motion in the bulk and two additional boundary conditions:

Euler–Lagrange equations with boundary

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} \right) &= 0 \quad \text{in } \mathcal{M}, \\ \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu y^i)} + \frac{\partial \ell^\mu}{\partial y^i} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial (\partial_\nu y^i)} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} P_\nu^\rho \right) \right) n_\mu &= 0 \quad \text{on } \partial\mathcal{M}, \\ \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} n_\mu n_\nu &= 0 \quad \text{on } \partial\mathcal{M}. \end{aligned} \quad (62)$$

5.3 Derivation of the Poincaré–Cartan form

Let us return again to the variational problem written in geometric form. The variation of the action given by the forms $\Theta_{\mathcal{M}}$ and $\Theta_{\partial\mathcal{M}}$ is given by equation (58). As we have already shown, if this variation is zero for every vector X , then the integrand must vanish. However, in the Lagrangian formalism we do not require this variation to be zero for every vector, but only for those that preserve holonomicity, that is, vectors of the form:

$$X_{hol} = \varepsilon^i \partial_{y^i} + D_\mu \varepsilon^i \partial_{y_\mu^i} \quad (63)$$

For generic variations of the action there was no problem in applying the fundamental lemma of the variational principle with boundary, since multiplying the variation vector X by an arbitrary function h produces another vector hX that is still an admissible vector within the generic variational principle.

However, in the Lagrangian formalism, multiplying by an arbitrary function hX_{hol} does not necessarily produce a holonomic vector. Therefore we cannot apply the fundamental lemma of the variational principle with boundary, since we need the integral to vanish for all possible functions h .

This problem has a geometric solution through the Poincaré–Cartan form. Indeed, although the set of admissible variations is too restrictive, the set of differential forms that define the same variational problem is very large. Looking at equation (51), we can deduce that two relative forms $(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}})$ and $(\Theta'_{\mathcal{M}}, \Theta'_{\partial\mathcal{M}})$ are equivalent if their pullback by $J^1\phi$ produces the same pair of differential forms. This is known as Lepage equivalence.

The idea is therefore to look for a pair of differential forms $(\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}})$ that are Lepage equivalent to the pair of Lagrangians $(\mathcal{L}d^n x, -\ell^\mu d^{n-1}x_\mu)$ and that directly encode the holonomicity of the system, in such a way that the variational principle is well defined for any vertical vector and not only for variations that preserve holonomicity.

To find the Poincaré–Cartan forms, both in the bulk and on the boundary, we can search for those that keep track of how integration by parts is performed in the variational principle of least action.

This can first be understood in the bulk case. The variation of the action is:

$$\begin{aligned}\delta S_{\mathcal{M}} &= \int_{\mathcal{M}} \frac{\partial \mathcal{L}}{\partial y^i} \varepsilon^i d^n x + \int_{\mathcal{M}} \left[d \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} \varepsilon^i d^{n-1} x_\mu \right) - \varepsilon^i \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} \right) d^n x_\mu \right] \\ &= \int_{\mathcal{M}} EL_i \varepsilon^i d^n x + \int_{\mathcal{M}} d \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} \varepsilon^i d^{n-1} x_\mu \right)\end{aligned}\quad (64)$$

so that the term in the second integral, after applying Stokes' theorem, does not contribute to the bulk but rather to the boundary. It is this second term that we consider important to retain in the variational principle, so we define the Poincaré–Cartan form as the form Θ_{PC} such that, when contracted with the vector $V = \varepsilon^i \partial_{y^i}$, it gives precisely the term that appears inside the total derivative. Therefore, the Poincaré–Cartan form must have the following structure:

$$\Theta_{PC} = \frac{\partial \mathcal{L}}{\partial y_\mu^i} dy^i \wedge d^{n-1} x_\mu + \text{other terms} \quad (65)$$

These other terms are adjusted so that Θ_{PC} is Lepage equivalent to the differential form $\mathcal{L} d^n x$, that is, so that they give rise to the same equations of motion.

It can be checked that the only form that is Lepage equivalent to $\mathcal{L} d^n x$ and has the structure described in the previous paragraph is:

$$\Theta_{PC} = \frac{\partial \mathcal{L}}{\partial y_\mu^i} dy^i \wedge d^{n-1} x_\mu - \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} y_\mu^i - \mathcal{L} \right) d^n x \quad (66)$$

Another way to write it is:

$$\Theta_{PC} = \frac{\partial \mathcal{L}}{\partial y_\mu^i} (dy^i - y_\nu^i dx^\nu) \wedge d^{n-1} x_\mu + \mathcal{L} d^n x \quad (67)$$

where the Lepage equivalence can be clearly seen.

We can do the same with the variation on the boundary:

$$\begin{aligned}\delta S_{\partial \mathcal{M}} &= \int_{\partial \mathcal{M}} \left(\frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i + \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} \partial_\nu \varepsilon^i \right) d^{n-1} x_\mu \\ &= \int_{\partial \mathcal{M}} \left(\frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i + \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} (\partial_\nu \varepsilon^i - n_\nu n^\rho \partial_\rho \varepsilon^i) \right) d^{n-1} x_\mu \\ &= \int_{\partial \mathcal{M}} \left(\frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i + \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} P_\nu^\rho \partial_\rho \varepsilon^i \right) d^{n-1} x_\mu \\ &= \int_{\partial \mathcal{M}} \frac{\partial \ell^\mu}{\partial y^i} \varepsilon^i d^{n-1} x_\mu - \int_{\partial \mathcal{M}} d \left(\frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} P_\nu^\rho \varepsilon^i d^{n-2} x_{\mu\rho} \right) \\ &\quad + \int_{\partial \mathcal{M}} \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial (\partial_\nu y^i)} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} P_\nu^\rho \right) \varepsilon^i d^{n-1} x_\mu \\ &= \int_{\partial \mathcal{M}} EL_i^\mu \varepsilon^i d^{n-1} x_\mu - \int_{\partial \mathcal{M}} d \left(\frac{\partial \ell^\mu}{\partial (\partial_\nu y^i)} P_\nu^\rho \varepsilon^i d^{n-2} x_{\mu\rho} \right)\end{aligned}\quad (68)$$

As in the previous case, we search for the Poincaré–Cartan form on the boundary such that contracting with the vector $V = \varepsilon^i \partial_{y^i}$ yields the term inside the exterior derivative of the second integral:

$$\Theta_{\partial \mathcal{M}} = \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho dy^i \wedge d^{n-2} x_{\mu\rho} + \text{other terms} \quad (69)$$

where, again, these other terms are chosen so that the Poincaré–Cartan form is Lepage equivalent to $-\ell^\mu d^{n-1} x_\mu$. The minus sign appears because we have defined the integral $\int_{(\mathcal{M}, \partial \mathcal{M})} (\alpha, \beta) = \int_{\mathcal{M}} \alpha - \int_{\partial \mathcal{M}} \beta$.

It can be checked that the following differential form is Lepage equivalent to $-\ell^\mu d^{n-1} x_\mu$ and contains the desired term coming from the integration by parts:

$$\Theta_{\partial \mathcal{M}} = \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho dy^i \wedge d^{n-2} x_{\mu\rho} - \left(\left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i + \ell^\mu \right) d^{n-1} x_\mu \quad (70)$$

Finally, the relative Poincaré–Cartan form, whose relative exterior derivative is the multisymplectic form for systems with boundary, is:

Poincaré–Cartan form with boundary

$$\Theta_{PC} = \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} dy^i \wedge d^{n-1}x_\mu - \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} y_\mu^i - \mathcal{L} \right) d^n x, \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho dy^i \wedge d^{n-2}x_{\mu\rho} + \left(\frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho y_\rho^i - \ell^\mu \right) d^{n-1}x_\mu \right) \quad (71)$$

The multisymplectic form is naturally given by the exterior derivative of the Poincaré–Cartan form:

$$\Omega = d\Theta_{PC} \quad (72)$$

5.4 Equations of motion via the Poincaré–Cartan form

With the forms $\Theta = (\mathcal{L}d^n x, -\ell^\mu d^{n-1}x_\mu)$ one does not obtain the correct equations of motion, because the admissible variations are not being encoded. By contrast, with the construction of the Poincaré–Cartan form we convert the variational problem with holonomic variations into a variational problem with general variations, so the correct equations of motion are those given by equations ?? presented in section ??.

The Poincaré form is specifically constructed so that the equations of motion are obtained from contracting the multisymplectic form with any vertical vector, not only those that preserve holonomicity. In this section we will check this explicitly. The relative Poincaré–Cartan form is:

$$\begin{aligned} \Theta_{PC} &= (\Theta_{\mathcal{M}}, \Theta_{\partial\mathcal{M}}) \\ &= \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} dy^i \wedge d^{n-1}x_\mu - \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} y_\mu^i - \mathcal{L} \right) d^n x, \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho dy^i \wedge d^{n-2}x_{\mu\rho} - \left(\left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i + \ell^\mu \right) d^{n-1}x_\mu \right) \end{aligned} \quad (73)$$

The multisymplectic form, obtained by taking the relative exterior derivative, is:

$$\begin{aligned} \Omega &= d\Theta_{PC} \\ &= (d\Theta_{\mathcal{M}}, d\Theta_{\partial\mathcal{M}}) \\ &= \left(\frac{\partial^2 \mathcal{L}}{\partial y_\mu^j \partial y_\mu^i} dy^j \wedge dy^i \wedge d^{n-1}x_\mu + \frac{\partial^2 \mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} dy_\nu^j \wedge dy^i \wedge d^{n-1}x_\mu - \frac{\partial^2 \mathcal{L}}{\partial x^\mu \partial y_\mu^i} dy^i \wedge d^n x - \frac{\partial^2 \mathcal{L}}{\partial y_\mu^j \partial y_\mu^i} y_\mu^i dy^j \wedge d^n x \right. \\ &\quad - \frac{\partial^2 \mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} y_\mu^i dy_\nu^j \wedge d^n x + \frac{\partial \mathcal{L}}{\partial y^i} dy^i \wedge d^n x, \frac{\partial \mathcal{L}}{\partial y_\mu^i} dy^i \wedge d^{n-1}x_\mu - \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y_\nu^j} P_\nu^\rho dy^j \wedge dy^i \wedge d^{n-2}x_{\mu\rho} \\ &\quad - \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho dy_\sigma^j \wedge dy^i \wedge d^{n-2}x_{\mu\rho} + \frac{\partial}{\partial x^\rho} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) dy^i \wedge d^{n-1}x_\mu \\ &\quad + \frac{\partial}{\partial y^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i dy^j \wedge d^{n-1}x_\mu + \frac{\partial}{\partial y_\sigma^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i dy_\sigma^j \wedge d^{n-1}x_\mu \\ &\quad \left. + \frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu dy_\rho^i \wedge d^{n-1}x_\mu + \frac{\partial \ell^\mu}{\partial y_\nu^i} n^\rho n_\nu dy_\rho^i \wedge d^{n-1}x_\mu + \frac{\partial \ell^\mu}{\partial y^i} dy^i \wedge d^{n-1}x_\mu \right) \end{aligned} \quad (74)$$

Contracting with a vector of the form $Z = Z_\mu^i \partial_{y_\mu^i}$ yields the following equations of motion:

$$\begin{aligned} \iota_Z \Omega &= \left(\frac{\partial^2 \mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} Z_\nu^j dy^i \wedge d^{n-1}x_\mu - \frac{\partial^2 \mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} y_\mu^i Z_\nu^j d^n x, \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho Z_\sigma^j dy^i \wedge d^{n-2}x_{\mu\rho} \right. \\ &\quad \left. - \frac{\partial}{\partial y_\sigma^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i Z_\sigma^j d^{n-1}x_\mu - \frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu Z_\rho^i d^{n-1}x_\mu + \frac{\partial \ell^\mu}{\partial y_\nu^i} n^\rho n_\nu Z_\rho^i d^{n-1}x_\mu \right) \end{aligned} \quad (75)$$

Evaluating the contraction on a section $J^1\phi$ we obtain:

$$\begin{aligned}
(J^1\phi)^*\iota_Z\Omega &= \left(\frac{\partial^2\mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} (\partial_\mu y^i - y_\mu^i) Z_\nu^j d^n x, \frac{\partial^2\ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho Z_\sigma^j \partial_\lambda y^i (\delta_\mu^\lambda d^{n-1} x_\rho - \delta_\rho^\lambda d^{n-1} x_\mu) \right. \\
&\quad \left. - \frac{\partial}{\partial y_\sigma^i} \left(\frac{\partial\ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial\ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i Z_\sigma^j d^{n-1} x_\mu - \frac{\partial\ell^\rho}{\partial y_\nu^i} P_\nu^\mu Z_\rho^i d^{n-1} x_\mu \right) \\
&= \left(\frac{\partial^2\mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} (\partial_\mu y^i - y_\mu^i) Z_\nu^j d^n x, \left(\left(\frac{\partial^2\ell^\rho}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\mu - \frac{\partial^2\ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho \right) Z_\sigma^j (\partial_\rho y^i - y_\rho^i) - \frac{\partial\ell^\rho}{\partial y_\nu^i} P_\nu^\mu Z_\rho^i + \frac{\partial\ell^\mu}{\partial y_\nu^i} n_\nu n^\rho Z_\rho^i \right) d^{n-1} x_\mu \right)
\end{aligned} \tag{76}$$

By the fundamental lemma of the calculus of variations with boundary, both differential forms must vanish on the boundary. Again, note that on the boundary this does not mean that all of their components are zero, but only their pullback by the inclusion, which in practice means contraction with n_μ :

$$(J^1\phi)^*\iota_Z\Omega = \left(\frac{\partial^2\mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} (\partial_\mu y^i - y_\mu^i) Z_\nu^j d^n x, \left(\left(\frac{\partial^2\ell^\rho}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\mu - \frac{\partial^2\ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho \right) Z_\sigma^j (\partial_\rho y^i - y_\rho^i) - \frac{\partial\ell^\rho}{\partial y_\nu^i} P_\nu^\mu Z_\rho^i + \frac{\partial\ell^\mu}{\partial y_\nu^i} n_\nu n^\rho Z_\rho^i \right) n_\mu d^{n-1} \tilde{x} \right) \tag{77}$$

But the penultimate term is proportional to $P_\nu^\mu n_\mu$, which is zero, so finally:

$$(J^1\phi)^*\iota_Z\Omega = \left(\frac{\partial^2\mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} (\partial_\mu y^i - y_\mu^i) Z_\nu^j d^n x, \left(\left(\frac{\partial^2\ell^\rho}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\mu - \frac{\partial^2\ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho \right) Z_\sigma^j (\partial_\rho y^i - y_\rho^i) + \frac{\partial\ell^\mu}{\partial y_\nu^i} n_\nu n^\rho Z_\rho^i \right) n_\mu d^{n-1} \tilde{x} \right) \tag{78}$$

In the bulk, the equations of motion yield:

$$\frac{\partial^2\mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} = 0 \tag{79}$$

In the regular case, the Hessian of the Lagrangian is nondegenerate and the above equation implies that the velocities are given by the derivatives of the fields:

$$y_\mu^i = \partial_\mu y^i \tag{80}$$

Since the section $j^1\phi$ is continuous, $y_\mu^i = \partial_\mu y^i$ also holds on the boundary.

In what follows, to derive the remaining equations of motion from the Poincaré–Cartan form, we will assume the Lagrangian is regular. If it is not, it is also possible to impose the condition $y_\mu^i = \partial_\mu y^i$ by restricting to holonomic sections only. While restricting to holonomic sections was a problem when using the forms $\mathcal{L}d^n x$ and $\ell^\mu d^{n-1} x_\mu$, it is not for the Poincaré–Cartan form, since the same equations of motion are obtained.

On the boundary, the equations of motion imply that the following quantity:

$$\left(\left(\frac{\partial^2\ell^\rho}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\mu - \frac{\partial^2\ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho \right) Z_\sigma^j (\partial_\rho y^i - y_\rho^i) + \frac{\partial\ell^\mu}{\partial y_\nu^i} n_\nu n^\rho Z_\rho^i \right) n_\mu = 0 \tag{81}$$

must vanish for every vector Z_ρ^i .

But, using the equation $y_\rho^i = \partial_\rho y^i$ on the boundary, the first term cancels and we arrive at:

$$\frac{\partial\ell^\mu}{\partial y_\nu^i} n_\nu n_\mu n^\rho Z_\rho^i = 0 \quad \forall Z_\rho^i \tag{82}$$

This implies that either $n^\rho = 0$ or the remaining factor is zero. At points of the boundary where the metric is nondegenerate, the normal vector n^ρ to the hypersurface is nonzero, hence we recover one of the boundary conditions:

$$\frac{\partial\ell^\mu}{\partial y_\nu^i} n_\nu n_\mu = 0 \tag{83}$$

Returning now to equation (74), we contract with $X = X^i \partial_{y^i}$ to find the rest of the equations of motion:

$$\begin{aligned}
\iota_X \Omega = & \left(\frac{\partial^2 \mathcal{L}}{\partial y^j \partial y_\mu^i} X^j dy^i \wedge d^{n-1} x_\mu - \frac{\partial^2 \mathcal{L}}{\partial y^j \partial y_\mu^i} X^i dy^j \wedge d^{n-1} x_\mu - \frac{\partial^2 \mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} X^i dy_\nu^j \wedge d^{n-1} x_\mu - \frac{\partial^2 \mathcal{L}}{\partial x^\mu \partial y_\mu^i} X^i d^n x \right. \\
& - \frac{\partial^2 \mathcal{L}}{\partial y^j \partial y_\mu^i} y_\mu^i X^j d^n x + \frac{\partial \mathcal{L}}{\partial y^i} X^i d^n x, - \frac{\partial \mathcal{L}}{\partial y_\mu^i} X^i d^{n-1} x_\mu + \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y^j} P_\nu^\rho X^j dy^i \wedge d^{n-2} x_{\mu\rho} \\
& - \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y^j} P_\nu^\rho X^i dy^j \wedge d^{n-2} x_{\mu\rho} - \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y_\sigma^j} P_\nu^\rho X^i dy_\sigma^j \wedge d^{n-2} x_{\mu\rho} - \frac{\partial}{\partial x^\rho} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) X^i d^{n-1} x_\mu \\
& \left. - \frac{\partial}{\partial y^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i X^j d^{n-1} x_\mu - \frac{\partial \ell^\mu}{\partial y^i} X^i d^{n-1} x_\mu \right)
\end{aligned} \tag{84}$$

Taking the pullback by a section:

$$\begin{aligned}
(J^1 \phi)^* \iota_X \Omega = & \left(\left(\frac{\partial^2 \mathcal{L}}{\partial y^j \partial y_\mu^i} (\partial_\mu y^i - y_\mu^i) X^j + \frac{\partial \mathcal{L}}{\partial y^i} X^i - \left(\frac{\partial^2 \mathcal{L}}{\partial x^\mu \partial y_\mu^i} + \frac{\partial^2 \mathcal{L}}{\partial y^j \partial y_\mu^i} \partial_\mu y^j + \frac{\partial^2 \mathcal{L}}{\partial y_\mu^i \partial y_\nu^j} \partial_\mu y_\nu^j \right) X^i \right) d^n x, \right. \\
& - \frac{\partial \mathcal{L}}{\partial y_\mu^i} X^i d^{n-1} x_\mu + \left(\frac{\partial^2 \ell^\rho}{\partial y_\nu^i \partial y^j} P_\nu^\mu - \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y^j} P_\nu^\rho \right) X^j \partial_\rho y^i d^{n-1} x_\mu - \frac{\partial}{\partial y^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) X^i \partial_\rho y^j d^{n-1} x_\mu \\
& - \frac{\partial}{\partial y_\sigma^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) X^i \partial_\rho y_\sigma^j d^{n-1} x_\mu - \frac{\partial}{\partial x^\rho} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) X^i d^{n-1} x_\mu \\
& \left. - \frac{\partial}{\partial y^j} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) y_\rho^i X^j d^{n-1} x_\mu - \frac{\partial \ell^\mu}{\partial y^i} X^i d^{n-1} x_\mu \right)
\end{aligned} \tag{85}$$

Using the derivative along sections $\frac{d}{dx^\rho} = \frac{\partial}{\partial x^\rho} + \partial_\rho y^j \frac{\partial}{\partial y^j} + \partial_\rho y_\sigma^j \frac{\partial}{\partial y_\sigma^j}$ we can simplify the expression somewhat:

$$\begin{aligned}
(J^1 \phi)^* \iota_X \Omega = & \left(\left(\frac{\partial^2 \mathcal{L}}{\partial y^j \partial y_\mu^i} (\partial_\mu y^i - y_\mu^i) X^j + \left(\frac{\partial \mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} \right) \right) X^i \right) d^n x, \right. \\
& \left(\left(\frac{\partial^2 \ell^\rho}{\partial y_\nu^i \partial y^j} P_\nu^\mu - \frac{\partial^2 \ell^\mu}{\partial y_\nu^i \partial y^j} P_\nu^\rho \right) (\partial_\rho y^i - y_\rho^i) X^j - \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} + \frac{\partial \ell^\mu}{\partial y^i} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) \right) X^i \right) d^{n-1} x_\mu \Big)
\end{aligned} \tag{86}$$

In the regular case, where $y_\rho^i = \partial_\rho y^i$, the left-hand terms cancel, so we finally obtain the Euler-Lagrange equations:

$$(J^1 \phi)^* \iota_X \Omega = \left(\left(\frac{\partial \mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} \right) \right) X^i d^n x, - \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} + \frac{\partial \ell^\mu}{\partial y^i} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) \right) X^i d^{n-1} x_\mu \right) \tag{87}$$

By the fundamental lemma of the calculus of variations with boundary, if these two differential forms are zero for every vector X^i , the equations of motion follow:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial y^i} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} \right) &= 0 \quad \text{in } \mathcal{M} \\
\left(\frac{\partial \mathcal{L}}{\partial y_\mu^i} + \frac{\partial \ell^\mu}{\partial y^i} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial y_\nu^i} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial y_\nu^i} P_\nu^\rho \right) \right) n_\mu &= 0 \quad \text{on } \partial \mathcal{M}
\end{aligned} \tag{88}$$

6 Examples

6.1 Scalar field (Neumann)

For Neumann boundary conditions, we search that the boundary equations of motion are $n^\mu \partial_\mu \phi = 0$. For this, we can use the following lagrangians in the bulk and the boundary:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad \ell^\mu = 0 \quad (89)$$

Using the equations of motion in the bulk and on the boundary:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) &= 0 \quad \text{in } \mathcal{M}, \\ \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} + \frac{\partial \ell^\mu}{\partial \phi} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial (\partial_\nu \phi)} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial (\partial_\nu \phi)} P_\nu^\rho \right) \right) n_\mu &= 0 \quad \text{on } \partial \mathcal{M}, \\ \frac{\partial \ell^\mu}{\partial (\partial_\nu \phi)} n_\mu n_\nu &= 0 \quad \text{on } \partial \mathcal{M}. \end{aligned} \quad (90)$$

The equations of motion for these lagrangians read:

$$\begin{aligned} \mathcal{M} : \partial_\mu \partial^\mu \phi &= -V'(\phi) \\ \partial \mathcal{M} : n_\mu \partial^\mu \phi &= 0 \\ \partial \mathcal{M} : 0 &= 0 \end{aligned} \quad (91)$$

Otherwise, if we don't want to fix the derivative to be zero and we want to fix it to be proportional to some function f^μ , we can change the lagrangian on the boundary:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad \ell^\mu = -\phi f^\mu \quad (92)$$

The equations of motion for these lagrangians read:

$$\begin{aligned} \mathcal{M} : \partial_\mu \partial^\mu \phi &= -V'(\phi) \\ \partial \mathcal{M} : n_\mu \partial^\mu \phi &= n_\mu f^\mu \\ \partial \mathcal{M} : 0 &= 0 \end{aligned} \quad (93)$$

For the case $\ell^\mu = 0$, fixing the normal derivative to be zero, we have the following Poincaré-Cartan form:

$$\Theta_{PC} = \left(\phi^\mu d\phi \wedge d^{n-1}x_\mu - \left(\frac{1}{2} \phi_\mu \phi^\mu + V(\phi) \right) d^n x, 0 \right) \quad (94)$$

The multisymplectic form:

$$\Omega = \underline{d}\Theta_{PC} = (d\phi^\mu \wedge d\phi \wedge d^{n-1}x_\mu - \phi^\mu d\phi_\mu \wedge d^n x - V'(\phi) d\phi \wedge d^n x, \phi^\mu d\phi \wedge d^{n-1}x_\mu) \quad (95)$$

If $V'(\phi) = 0$, the vector ∂_ϕ is hamiltonian and has the following associated relative hamiltonian form:

$$F = (F_{\mathcal{M}}, F_{\partial \mathcal{M}}) = (-\phi^\mu d^{n-1}x_\mu, 0) \quad (96)$$

For constant vectors $\xi = -\xi^\mu \partial_\mu$, the vector ξ is hamiltonian if and only if $\xi \cdot n = 0$ and it has as associated hamiltonian form (Not necessarily $V'(\phi) = 0$):

$$G_\xi = \left(-\xi^\mu \phi^\nu d\phi \wedge d^{n-2}x_{\mu\nu} + \xi^\mu \left(\frac{1}{2} \phi_\nu \phi^\nu + V(\phi) \right) d^{n-1}x_\mu, 0 \right) \quad (97)$$

This observable is closely related with the energy momentum tensor when taking the pullback by a section $j^1\phi$.

These two are conserved quantities.

6.2 Scalar field (Dirichlet)

For Dirichlet boundary conditions, we search that the boundary equations of motion are $n^\mu \partial_\mu \phi = 0$. For this, we can use the following lagrangians in the bulk and the boundary:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad \ell^\mu = -\phi \partial^\mu \phi \quad (98)$$

Using the equations of motion in the bulk and on the boundary:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) &= 0 \quad \text{in } \mathcal{M}, \\ \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} + \frac{\partial \ell^\mu}{\partial \phi} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial (\partial_\nu \phi)} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial (\partial_\nu \phi)} P_\nu^\rho \right) \right) n_\mu &= 0 \quad \text{on } \partial \mathcal{M}, \\ \frac{\partial \ell^\mu}{\partial (\partial_\nu \phi)} n_\mu n_\nu &= 0 \quad \text{on } \partial \mathcal{M}. \end{aligned} \quad (99)$$

The equations of motion for these lagrangians read:

$$\begin{aligned} \mathcal{M} : \partial_\mu \partial^\mu \phi &= -V'(\phi) \\ \partial \mathcal{M} : 0 &= 0 \\ \partial \mathcal{M} : \phi &= 0 \end{aligned} \quad (100)$$

Otherwise, if we don't want to fix the field to be zero and we want to fix it to be some function Φ , we can change the lagrangian on the boundary:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad \ell^\mu = -\phi \partial^\mu \phi + \phi \Phi \quad (101)$$

The equations of motion for these lagrangians read:

$$\begin{aligned} \mathcal{M} : \partial_\mu \partial^\mu \phi &= -V'(\phi) \\ \partial \mathcal{M} : 0 &= 0 \\ \partial \mathcal{M} : \phi &= \Phi \end{aligned} \quad (102)$$

For the case $\ell^\mu = -\phi \partial^\mu \phi$, fixing the field to be zero, we have the following Poincaré-Cartan form:

$$\Theta_{PC} = \left(\phi^\mu d\phi \wedge d^{n-1}x_\mu - \left(\frac{1}{2} \phi_\mu \phi^\mu + V(\phi) \right) d^n x, \phi \phi^\mu d^{n-1}x_\mu - \Phi \phi^\mu d^{n-1}x_\mu \right) \quad (103)$$

and the multisymplectic form:

$$\Omega = \underline{d}\Theta_{PC} = (d\phi^\mu \wedge d\phi \wedge d^{n-1}x_\mu - \phi^\mu d\phi_\mu \wedge d^n x - V'(\phi) d\phi \wedge d^n x, -\phi d\phi^\mu \wedge d^{n-1}x_\mu + \Phi d\phi^\mu \wedge d^{n-1}x_\mu) \quad (104)$$

For $\Phi = 0$, the constant vector fields of the form $\xi = \xi^\mu \partial_\mu$ are hamiltonian if and only if $\xi \cdot n = 0$ and it has associated the following relative hamiltonian form:

$$G_\xi = \left(\xi^\mu \phi^\nu d\phi \wedge d^{n-2}x_{\mu\nu} + \xi^\mu \left(\frac{1}{2} \phi_\nu \phi^\nu + V(\phi) \right) d^{n-1}x_\mu, \xi^\mu \phi \phi^\nu d^{n-2}x_{\mu\nu} \right) \quad (105)$$

6.3 Electromagnetism (Neumann)

The lagrangians in the bulk and on the boundary:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad \ell^\mu = 0 \quad (106)$$

The equations of motion:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A_\alpha} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\alpha)} \right) &= 0 \quad \text{in } \mathcal{M}, \\ \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\alpha)} + \frac{\partial \ell^\mu}{\partial A_\alpha} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial (\partial_\nu A_\alpha)} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial (\partial_\nu A_\alpha)} P_\nu^\rho \right) \right) n_\mu &= 0 \quad \text{on } \partial \mathcal{M}, \\ \frac{\partial \ell^\mu}{\partial (\partial_\nu A_\alpha)} n_\mu n_\nu &= 0 \quad \text{on } \partial \mathcal{M}. \end{aligned} \quad (107)$$

And substituting the lagrangians in the bulk and on the boundary:

$$\begin{aligned} \mathcal{M} : \partial_\mu F^{\mu\nu} &= 0 \\ \partial \mathcal{M} : n_\mu F^{\mu\nu} &= 0 \\ \partial \mathcal{M} : 0 &= 0 \end{aligned} \quad (108)$$

The Poincaré-Cartan form:

$$\Theta_{PC} = \left(-F^{\mu\nu} dA_\nu \wedge d^{n-1}x_\mu - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} d^n x, 0 \right) \quad (109)$$

And the multisymplectic form:

$$\Omega = -d\Theta_{PC} = \left(-dA_\nu \wedge dF^{\mu\nu} \wedge d^{n-1}x_\mu + \frac{1}{2} F_{\mu\nu} dF^{\mu\nu} \wedge d^n x, F^{\mu\nu} dA_\nu \wedge d^{n-1}x_\mu \right) \quad (110)$$

For constant vectors $\xi = \xi^\mu \partial_\mu$, the vector ξ is hamiltonian if and only if $\xi \cdot n = 0$ and it has the following associated relative hamiltonian form:

$$G_\xi = \left(-\xi^\rho F^{\mu\nu} dA_\mu \wedge d^{n-2}x_{\nu\rho} + \frac{1}{4} \xi^\rho F_{\mu\nu} F^{\mu\nu} d^{n-1}x_\rho, 0 \right) \quad (111)$$

The vector $V = \partial_\nu \varepsilon \partial_{A_\nu}$ is hamiltonian and has as associated relative form $F_\varepsilon = (F^{\mu\nu} \partial_\nu \varepsilon d^{n-1}x_\mu, 0)$, which is a conserved quantity.

6.4 Electromagnetism (Dirichlet)

The lagrangians in the bulk and on the boundary:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad \ell^\mu = A_\nu F^{\mu\nu} \quad (112)$$

The equations of motion:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A_\alpha} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\alpha)} \right) &= 0 \quad \text{in } \mathcal{M}, \\ \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\alpha)} + \frac{\partial \ell^\mu}{\partial A_\alpha} + \frac{d}{dx^\rho} \left(\frac{\partial \ell^\rho}{\partial (\partial_\nu A_\alpha)} P_\nu^\mu - \frac{\partial \ell^\mu}{\partial (\partial_\nu A_\alpha)} P_\nu^\rho \right) \right) n_\mu &= 0 \quad \text{on } \partial \mathcal{M}, \\ \frac{\partial \ell^\mu}{\partial (\partial_\nu A_\alpha)} n_\mu n_\nu &= 0 \quad \text{on } \partial \mathcal{M}. \end{aligned} \quad (113)$$

And substituting the lagrangians in the bulk and on the boundary:

$$\begin{aligned} \mathcal{M} : \partial_\mu F^{\mu\nu} &= 0 \\ \partial \mathcal{M} : \frac{d}{dx^\rho} (-\eta^{\rho\alpha} A^\nu P_\nu^\mu + \eta^{\mu\alpha} A^\nu P_\nu^\rho) n_\mu &= 0 \\ \partial \mathcal{M} : n^2 A_\rho &= (n \cdot A) n_\rho \end{aligned} \quad (114)$$

Operating the third equation we obtain $P_\nu^\mu A_\mu = 0$, while the second one does not add more information.

7 Conclusion

- Field theory in manifolds with boundaries can be described with multisymplectic formalism by changing the cohomology
- Boundary terms in the action encode the boundary conditions
- Dynamics of field theories in manifolds with boundaries are described by the Poincaré-Cartan relative form
- Outlook: second order lagrangians
- Edge and vertex terms changing the cohomology: $\underline{\underline{d}}(\alpha, \beta, \gamma, \dots) = (d\alpha, j^* \alpha - d\beta, (j \circ \kappa)^* \alpha - \kappa^* \beta - d\gamma, \dots)$